

Introduction

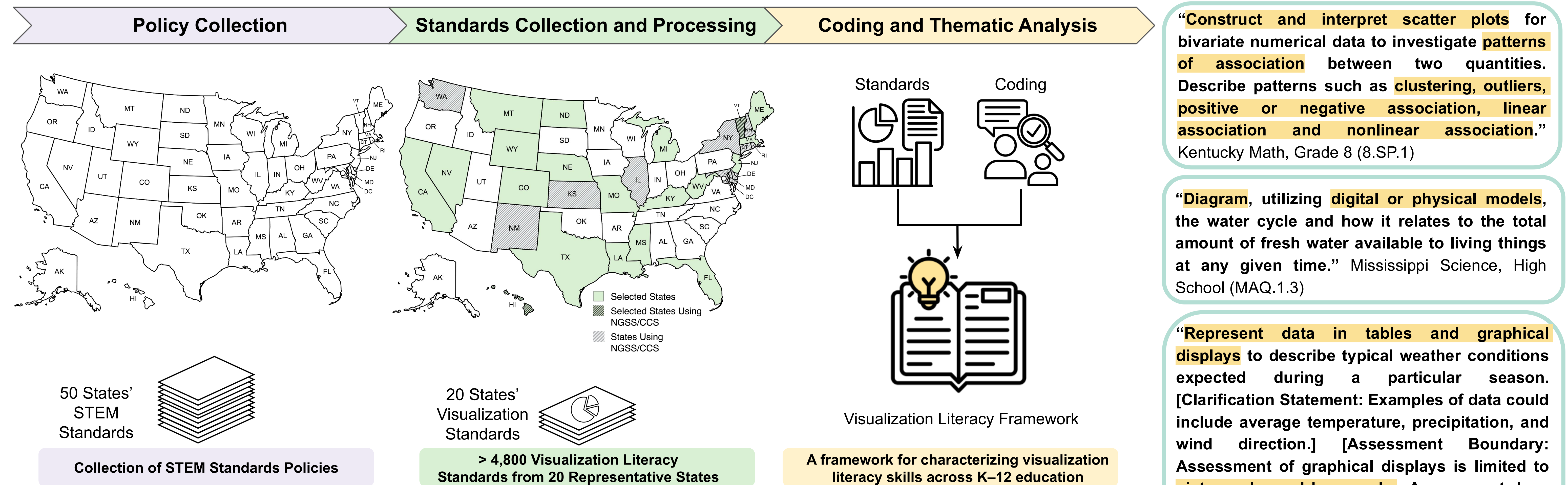
Prior work has built visualization literacy frameworks and assessments, but little is known about how these skills are taught in primary and secondary education (K–12). Knowing what visualization skills K–12 teaches helps researchers anticipate audience literacy and design assessments accordingly. We ask:

- What visualization literacy skills are students expected to learn throughout K–12 education?
- How do visualization tasks, idioms, and representation modes vary across grade levels and subjects?
- How do existing visualization literacy frameworks align with formal educational standards?

Contributions

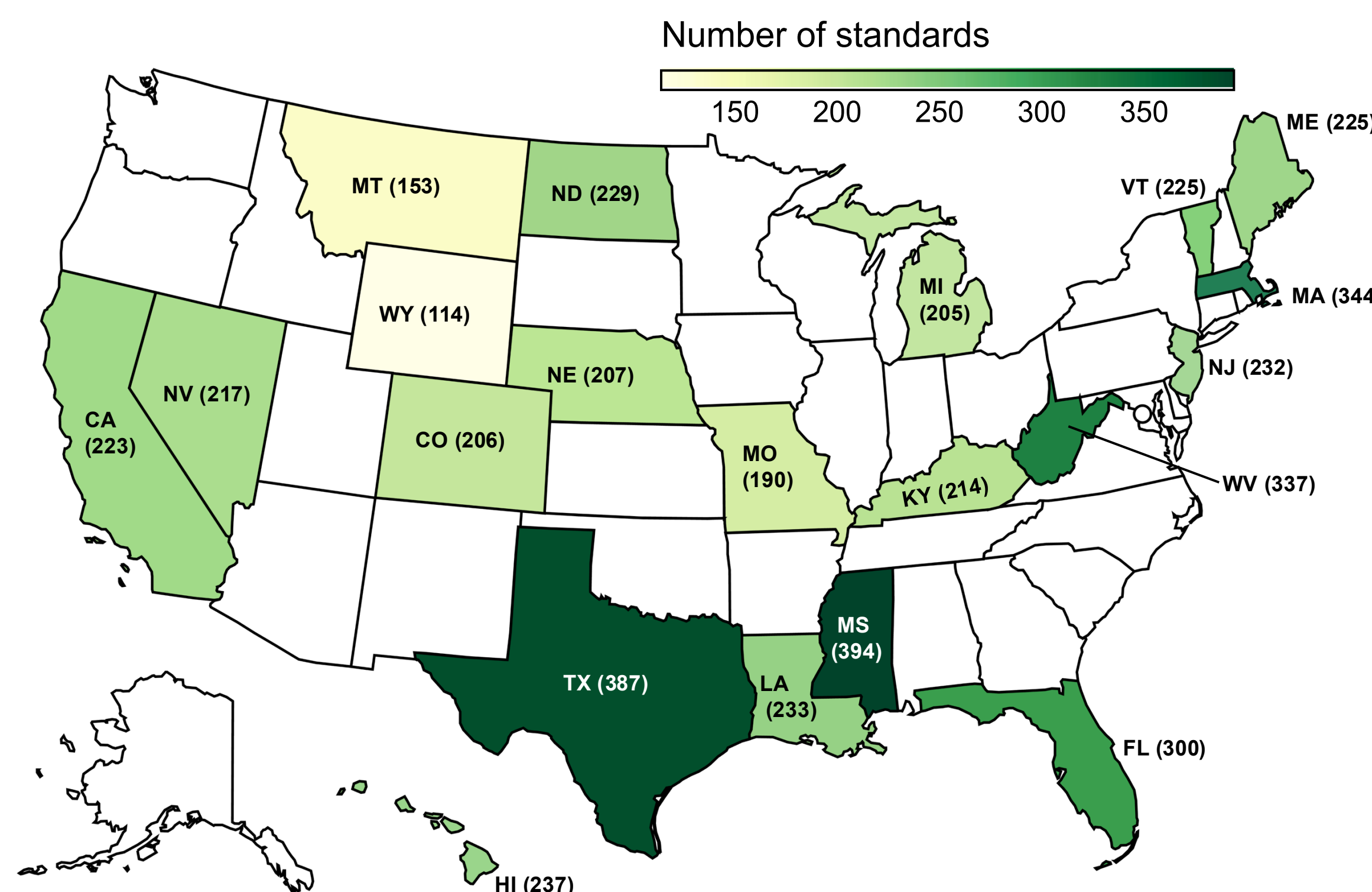
- A corpus of STEM education standards collected from all 50 U.S. states.
- A dataset of 4,872 visualization literacy standards extracted from 20 representative states' STEM education policy documents.
- An analysis of visualization tasks, idioms, and representation modes across grades and subjects.

Methodology

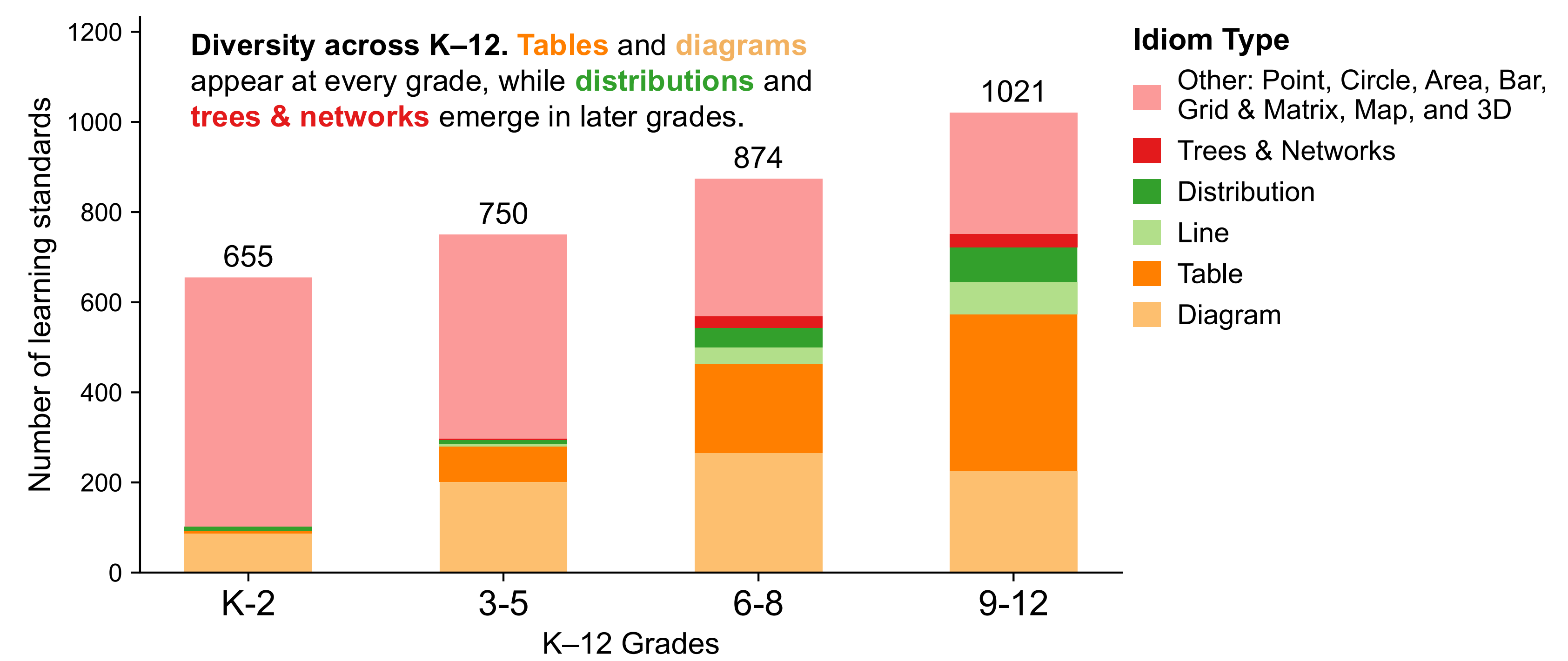


Results

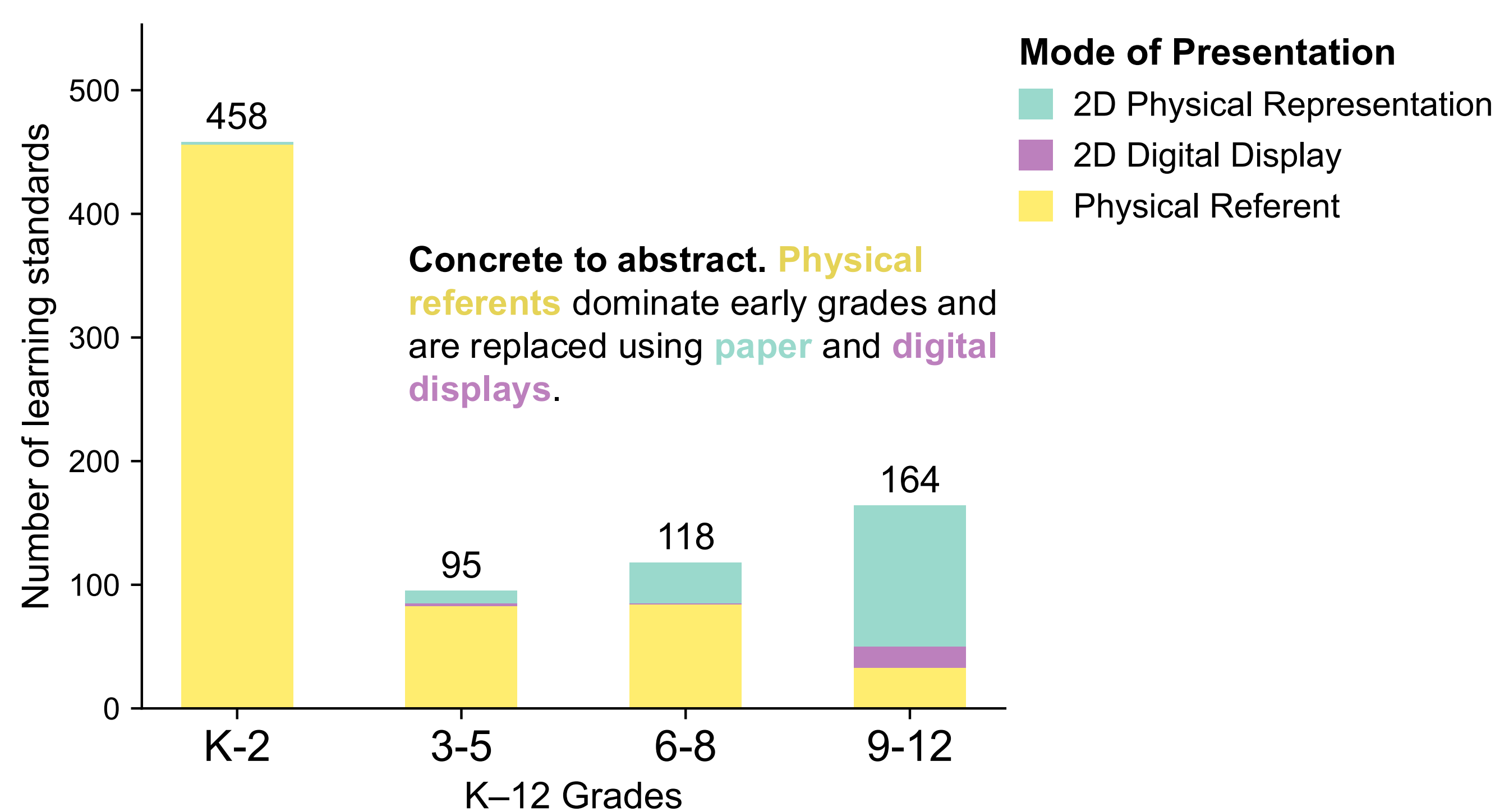
a) Visualization Literacy Standards by State



b) Visualization Idiom by Grade



c) Visualization Mode by Grade



d) VLAT Task by Grade

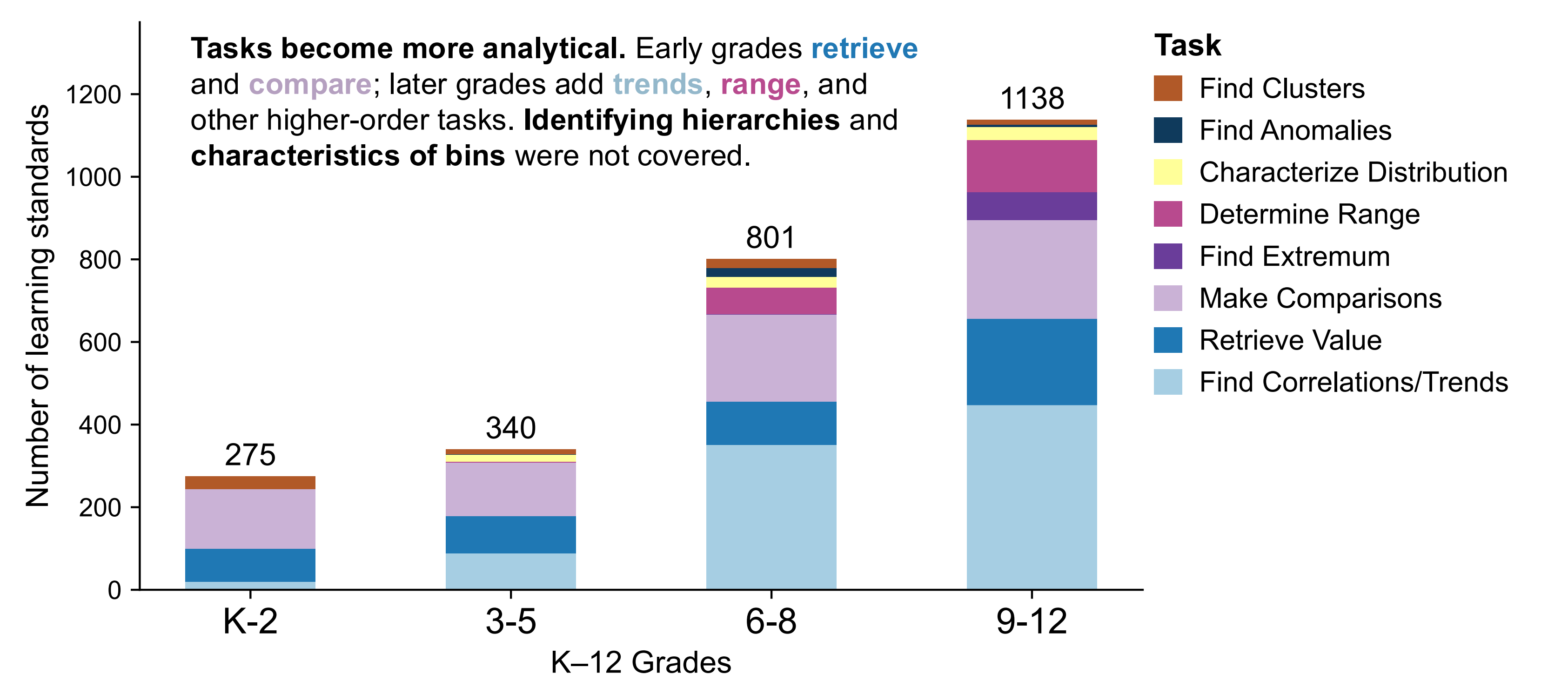


Figure 3: (a) Choropleth map of total visualization literacy standards by state; stacked bar charts showing the (b) distribution of visualization idioms³, (c) representation modes, and (d) VLAT tasks⁴ across K–12.

Future Works

- Develop a K–12 visualization literacy framework.
- Interview educators to learn how standards translate into teaching practice.
- Extend the corpus to non-STEM subjects, AP standards, and district-level cases.

Acknowledgments and References

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- [1] A. Bryman and B. Burgess, Eds., *Analyzing Qualitative Data*, 1st ed. London, UK: Routledge, 1994, doi:10.4324/9780203413081.
- [2] T. Munzner. Chapter 5 Marks and Channels. In *Visualization analysis & design*, A K Peters visualization series, pp. 95–115. 1st edition ed., 2014.
- [3] M. A. Borkin, A. A. Vo, Z. Bylinskii, P. Isola, S. Sunkavalli, A. Oliva, and H. Pfister, "What makes a visualization memorable?," *IEEE TVCG*, vol. 19, no. 12, pp. 2306–2315, Dec. 2013, doi:10.1109/TVCG.2013.234.
- [4] S. Lee, S.-H. Kim, and B. C. Kwon, "VLAT: Development of a Visualization Literacy Assessment Test," *IEEE TVCG*, vol. 23, no. 1, pp. 551–560, Jan. 2017, doi:10.1109/TVCG.2016.2598920.



Scan for dataset